Republic of the Philippines
PARTIDO STATE UNIVERSITY
COLLEGE OF ENGINEERING AND COMPUTATIONAL SCIENCES
Department of Computational Sciences | Bachelor of Science in Information Technology
Goa, Camarines Sur

BOOK OF ABSTRACT (eCOPY) — GROUP COVER & IMRAD QR ACCESS

"AURORA: ACADEMIC UNIFIED REVIEW, OBSERVATION, RATING, AND ASSESSMENT — A PAPERLESS ACADEMIC DEFENSE WORKFLOW & QUALITY ASSURANCE PLATFORM"

SCAN FOR FULL IMRAD PAPER (PDF)

ACM SIGPLAN Proceedings Format • ParSU CECS



Direct Online PDF Access Link:
https://aurora-parsu.vercel.app/imrad/AURORA_IMRAD_Paper.pdf

Proponents: Mañago, G. V. • Castillo, C. A. • Buenafe, R. M. • Encinas, L. M.
Research Adviser: Prof. Froiland B. Peñalosa

Proponents / Researchers	Gene Vincent Mañago • Carol Ann Castillo • Rona Mae Buenafe • Lara Mae Encinas
Degree & Specialization	Bachelor of Science in Information Technology (BSIT)
Research Adviser	Prof. Froiland B. Peñalosa
Capstone Coordinator & Chair	Dr. Kennedy C. Cuya
College Dean	Engr. Emelina R. Padayao
IMRAD PDF Online URL	https://aurora-parsu.vercel.app/imrad/AURORA_IMRAD_Paper.pdf

EXECUTIVE CAPSTONE ABSTRACT

Traditional undergraduate capstone and thesis defense workflows at Partido State University rely on physical manuscript printing, manual routing of Form DCS-CF-03 endorsement slips, spreadsheet-based room scheduling, and hand-computed Form DCS-CF-04 evaluation sheets. These manual procedures cause prolonged review turnaround times (averaging 18.4 days), scheduling collisions, recurring printing expenses for students, and fragile paper archives.

This study developed AURORA (Academic Unified Review, Observation, Rating, and Assessment), a cloud-native, paperless academic defense workflow and quality assurance platform engineered for the College of Engineering and Computational Sciences (CECS). Developed using the Agile Scrum methodology across Academic Year 2025–2026 through the 1st Semester of Academic Year 2026–2027 (August 2025 to October 2026) with Next.js 16, React 19, TypeScript, and Supabase PostgreSQL 15, AURORA integrates four core modules: (1) SHA-256 versioned PDF manuscript management with college-customizable defense stages and Form DCS-CF-03 adviser gating; (2) a dual-pane split-screen PDF.js review workspace with real-time WebSocket coordinate annotations; (3) an automated defense scheduler with room-clash detection and an academic Conflict-of-Interest Firewall; and (4) a dynamic weighted rubric engine (Form DCS-CF-04) secured by Republic Act No. 8792 step-up password re-authentication, tamper-evident certificate serials (AURORA-YYYY-NNNNNN), and a 3-2-1 Disaster Recovery & Self-Healing Maintenance Console.

Empirical evaluation with N = 27 ParSU CECS stakeholders (15 BSIT students, 5 research advisers, 5 defense panelists, 1 capstone coordinator, and 1 college dean) using a validated 10-item System Usability Scale (SUS) and 27-item UTAUT questionnaire (Good & Scates validity = 4.89/5.00; Cronbach's $\alpha = 0.946$) demonstrated a Composite SUS Score of 91.8 / 100 (Grade A — Excellent) and a Composite UTAUT Grand Mean of 4.86 / 5.00 (Highly Acceptable), while reducing manuscript review turnaround time by 85.9% (from 18.4 days to 2.6 days).

Keywords: *Paperless Defense Workflow, Coordinate PDF Annotation, Republic Act No. 8792 Digital Signature, Row-Level Security, System Usability Scale (SUS), UTAUT.*